Transportation Model

Basically in transportation there are 4 Right (or Principle) which you must ensure.

1) Right quality: what makes a quality product is different from other product
2) Right quantity: It is not such sufficient that the quality is right but you must make sure quantity is produced & available. If not it will lead to customer dissatisfaction. The effect is that its lead to bad reputation. Due to somebody will be negative advertising against you
3) At the right time: eg A. The product must be available when it is needed otherwise it will become no longer valuable.
4) Right manufacturing cost: By extension this include the cost of making the Product to be available in market at a reasonable cost. This 4th right take us to the concept of "Transportation". Thus you must ensure an optimum transportation cost as its is necessary that you can sell below the cost of the product + transportation cost + all other expenses

NB Stakeholder is a person or group of person who has stake in something and has an expectation from it

Transportation model is a process of ensuring an optimum physical distribution of goods from various sources to various destination. The sources are the supply point & the destination are the demand point.

Solution Procedure to Transportation Model

There are 2 solution procedure to transportation model
Initial basic feasible solution
Optimum solution

Under Initial basic feasible solution we have 3 method
1) North-West Corner method
2) least-cost method

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3. Vojel approximation method

Under Optimum Solution we have 2 method
1. Stepping stone algorithm method
2. Modified distribution method

NB To solve a Transportation Problem, it must be a Balanced Transportation Model

A Balanced transportation model exist where the total demand is equal to total supply or Vice-Versa, i.e what you are going to produce is binding on the market

Unbalanced transportation model exist when the total demand is greater than total supply or Vice-Versa
In reality we don't have balanced model therefore there is a need to balance it

To solve an unbalanced transportation model a dummy row or column will be introduced to absorb the difference between demand & supply

Illustration

XYZ Nigeria Ltd transport product from 3 factories to 4 warehouses of the customer. The supply capacities of the factories cost of transportation as well as demand of the 4 warehouses are presented in the table below

From \ To	D_1	D_2	D_3	D_4	Supply
S_1	190	300	500	100	700
S_2	700	300	400	600	900
S_3	400	80	700	200	1800
Demand	500	800	900	1400	3400 / 3400

Several route ($S_1 - S_3$) to take decision
Cell in with X which shows at every Unit transported from a source to any source is 190 though it can be greater than zero but not Negative If it is zero it means u didn't transport above

Required: 1. Formulate the Problem as a linear Programming model
2. Solve for initial basic feasible solution & determine the total transportation cost of it

Transportation model is a family model or special class of linear Programming model

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(Cb maximize or minimize)

Linear Programming model: is a model technique for optimization
There are 4 steps in formulating linear Programming model

1 Identify the decision variable: As manager you must know what you are taking decision on eg either to leave or to stay. This is due to as it is only the one within your range what determines the state of nature ~~is combination of~~ outcome of the problem is the and the combination of the problems

2 State the Objective function: Having known what you are making decision on you must know after gotten it what to do

3 State the constraints: Due to at any time you set an objective as a manager there are constraint; you don't assume all is well

4 You add the non-negativity function: As a management scientist you see every of your commitment as investment (ie sacrificing present consumption for higher future consumption) (NB the higher rate it difference from savings)

Translation cost by quantity

Back to the Question ⇒ First is to ensure your demand = supply
Thus add the supply column & demand row = 3400 both
Then identify the decision variable

	D_1	D_2	D_3	D_4	Supply
S_1	190	300	500	1600	700
	X	700	300	400	900
S_2		400	800	700	200	1800
S_3					
Demand	500	800	700	1400	

Step 1: Decision Variables

Minimize: $190X_{S_1D_1} + 300X_{S_1D_2} + 100X_{S_1D_3} + 700X_{S_2D_1} + 300X_{S_2D_2} + 400X_{S_2D_3} + 600X_{S_2D_4} + 400X_{S_3D_1} + 800X_{S_3D_2} + 700X_{S_3D_3} + 200X_{S_3D_4}$

X = the quantity that can be transmitted

X is unknown quantity that can be transported from A to B

The constraint are 2 which are supply & demand

So Subject to (S.T): $X_{S_1D_1} + X_{S_1D_2} + X_{S_1D_3} = 700$

+ constraints $X_{S_2D_1} + X_{S_2D_2} + X_{S_2D_3} = 900$

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Step 2: Stating the Objective Function

In practice maximise is different from minimize as you maximize profit, opportunities etc but in minimize you minimize cost, loss. Traditionally, a transportation model is a minimize case unless in Odega business it would be maximization in order to maximize the travel for return. In this case we will minimize.

We constraints 2 which are Demand & Supply

Subject to S-T $X_{S1}O_1 + X_{S1}O_2 + X_{S1}O_3 + X_{S1}O_4 = 700$ } Supply constraints
$X_{S2}O_1 + X_{S2}O_2 + X_{S2}O_3 + X_{S2}O_4 = 900$
$X_{S3}O_1 + X_{S3}O_2 + X_{S3}O_3 + X_{S3}O_4 = 1800$

$X_{O1}S_1 + X_{O1}S_2 + X_{O1}S_3 = 500$
$X_{O2}S_1 + X_{O2}S_2 + X_{O2}S_3 = 800$ } Demand constraints
$X_{O3}S_1 + X_{O3}S_2 + X_{O3}S_3 = 700$
$X_{O4}S_1 + X_{O4}S_2 + X_{O4}S_3 = 1400$

Step 4: Non-Negative Function $X_{ij} \geq 0$ For all i and j

i is row, j is column.

This means the quantity can be more or equal to zero but must/cannot be less. This however cannot be solved manually because of the too many variables.

Mr Gabriel Olumuyiwa ONDBOTO

2/3/2019

INVENTORY MANAGEMENT

Inventory is the stock of any item or resource used in any organization eg University of Lagos is a service industry. Just like Hospital offer services of nursing and drugs.

Inventory system is a set of policy & control to check how stock quantity level in order to prevent stock. It determine what levels shud be maintained when stock should be replenished & how large orders shud be.

Inventory is classify as raw material (source oil) & Packaging material (sachet nylon) in manufacturing sector. It also include finished goods, work in process, Component Parts.

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Purpose of Inventory

1) To maintain independence of operations to ensure flexibility in production
2) To meet variation in demand
3) To ensure flexibility in production scheduling
4) To provide a safeguard of variation in raw material delivery time
5) To take advantage of economic purchase order size
eg As increase in purchase order size the price of the product will increase reduce

Advantages of Inventory

1) It ensures continuity of supply of materials needed for production
2) Anticipation stock you need customer expectation needs
3) It prevents operational delay
4) It enhance regular supply of goods
5) It enhances saving in negotiation cost

Disadvantages

1) The stock that you buy & keep tie down capital
2) The risk of deterioration of goods & obsolescence of finished goods

INVENTORY COST

1) Holding or Carrying cost: this are cost like warehouse it is refers to cost of storage handling insurance, breakage pilferage
2) Set up or Production cost this include equipment set up charging & moving
3) Ordering cost: this is refers to managerial or purchase cost to meet up the requirement of raw order
4) Storage cost: Shortage cost

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Independent & Dependent Demand

The entire inventory system are unrelated to each other. Independent demand means the same item but can be purchased from different external sources eg For Cars you can purchase a Camary Car from USA or Japan

Dependent demand is relative straight forward how needed quantity of Dependent demand are calculated based on what it is demand eg If an Automobile is manufacturing 500 Cars it will depend on 5 tyres for each of the 500 Cars making 2500 tyres in total due to Spare Part

Inventory Ordering System

This Provide the organisational structure for controls monoitoring Stock

1) Ordering & receiving of goods
2) Keep track of what has been order eg who is the order coming from & tracking of the order
3) Placement timing the order Placement

There are basically 3 Ordering system for Inventory

1) Fixed order Quantity system
2) The Fixed order internal system (FOIS)
3) ABC inventory system analysis system

NB Most multinational companies choose ABC

1) Fixed order quantity system: Here, order are being placed on fixed quantity (Q) on each inventory cycle eg If your Q item is 6,000 & it got to 5,000 thus you have to re-order

Advantages

This system is appropriate when item of inventory are low value
No need for

Disadvantages

Too much Paper work
Demand Pattern may changes
Order are frequent & smaller

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(FOIS): Here, replenishment system are made in specified intervals. It can be weekly monthly or yearly

Advantages

It is useful when dealing with different items

Disadvantages

Operational cost are much higher

3) ABC: effective control of inventory can be costly & time consuming. Here they look at all the materials together then evaluate the value each of the materials then grade it perhaps 10-20% of the items (this raw material could be spare parts etc) eg taking 20% of 100,000 items that account for about 70-80% of the value i.e the most expensive of the item as classification A. Here you have taken care of most things

In Group B you take 30-40% of the item that account for 30-40% of the value of the item

In Group C you take 30-40% of the item that account of insignificant item

NB Stock out is criminal in industry as marketing department usually calculate loss of opportunity within the production stoppage

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SEQUENCING

Introduction

All things being equal, every organization will want to utilize its productive system effectively & efficiently so as to maximize profit by meeting delivery deadline.

Jobs involving many operations have to be performed & where there are limited resources in terms of plant & machineries on which the job have to be performed and this leads us to the idea of Sequencing

Definition of Sequencing

Sequencing concerned with the selecting of the appropriate order for a series of jobs to be done on a finite numbers of service facilities. The implication is that incase you have a series of jobs you need to apply appropriate method to select to meet up demand

Assumption of Sequencing

1) All machines can process only one job at a time
2) The processing time is known & fixed
3) When a job start it must be completed
4) Preceding operation must be completed before the succeeding are
5) All jobs are in a specified order as soon as possible
6) All jobs are ready for processing
7) All machines have different capabilities & capacities
8) Processing time of a job has no relation with the order in which the job is process
9) No time is wasted in shifting a job from one machine to another. The wasted time is zero

Types Of Sequencing Problem

Type 1 N-Jobs on one machine : where N is greater than 1
Type 2 N-Jobs on two machines
Type 3 N-Jobs on N machines where N is greater than 2

Sequencing Terminologies

A) Machine: is the facility which has some processing capability

B) Scheduling: This is when time of processing a job is specified
C) Total Elapsed time: This is the time that lapses between the starting of the first job & the finishing hour of the last job
D) Idle time: It is the time the facility or machine were not used during the elapse time eg Rest
E) No Passing Rule: It refers to the rule of maintaining the order in which jobs are to be processed on a given machine i.e a particular job must first be completed before any other one can start

Example: The Processing time for Jobs A, B, C, D, E and the due dates for these Jobs are given in the table below

Job	Processing Time	Due date (days)
A	9	16
B	7	20
C	5	25
D	8	20
E	6	40

The Jobs are to be sequenced according to:
i) First come first serve (FCFS)
ii) SPT (Shortest Processing Time) eg designing a wedding gown, and have a trouser to repair, you will repair the trouser first bcos it does not take time
iii) LS (Least slack): This is knowing the difference in doing the Job
iv) Expected Due Date (EDD)

Solution

FCFS

Step 1: Repeat the first-two columns

Job	Process Time
A	9
B	7
C	5
D	8
E	6

Step 2: Cummulate the Process time & add them together. (This is called the flow time.

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Job	Process Time	Flow Time
A	9	9
B	7	16
C	5	21
D	8	29
E	6	35
		110

Step 3: Reveal the due date

Job	Process time	Flow time	Due Date
A	9	9	16
B	7	16	20
C	5	21	25
D	8	29	20
E	6	35	40

Step 4: Determine the late jobs by checking the due date with the flow time. For Example, Job A was finished in 9 days as against the promise of delivery of delivery 16 days therefore is no lateness

2) Job B was finished in 16 days against the promise of delivery of 20 days hence there is no lateness.

If we check this for all the job we shall arrive at the following table

Job	Process Time	Flow Time	Due Date	Lateness
A	9	9	16	0
B	7	16	20	0
C	5	21	25	0
D	8	29	20	9
E	6	35	40	0
	35	110		9

Using FCFS

1) Total completion time is 35 days

2) Average completion time = Total time flow / No of Jobs = 110/5 = 22 days

3) Average number of work at the center = Total flow time / Total Processing time = 110/35 = 3.14 Jobs

4) Average Job lateness = Total lateness / No of Jobs = 9/5 = 1.8 days

5) Tardy Job (late job) = JOB D = 9

The shortest Processing time is JOB 6

Demand pat... order are frequent & smaller

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24/6/2019

LEAST SLACK

Step 1: The difference between Due date and Processing time give us the slack

Job	Due Date	Processing Time	Slack
A	16	9	7
B	20	7	13
C	25	5	20
D	20	8	12
E	40	6	34

Step 2: Arrange the Job Processing time and due date based on the ascending order of the slack as follows

Job	Processing Time	Slack
A	9	7
D	8	12
B	7	13
C	5	20
E	6	34

Step 3: Delete the Slack & cummulate the Processing time & Put the values in between Processed time & due date & tag it Flowtime as below

Job	Process Time	Flow Time	Due Date
A	9	9	16
D	8	17	20
B	7	24	20
C	5	29	25
E	6	35	40
	35	114	

Step 4: Determine the late Job by checking the due date with Flow time

JOB	Process Time	Flow Time	Due date	Lateness
A	9	9	16	0
D	8	17	20	0
B	7	24	20	4
C	5	29	25	4
E	6	35	40	0
	35	114		

Solution point: Test if it is notes that

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Using least Slack

1) Total completion time is 35 days
2) Average completion is equal to Total Flow time / No of Job = 115 / 5 = 22.8 days
3) Average number of works at the center = Total Flow time / Total Processing time = 115 / 35 = 3.28 jobs
4) Average Job lateness = Total lateness / No of Jobs = 8 / 5 = 1.46 days
5) Tardy Jobs i.e Job lateness = 2 Jobs which are Job B & Job C

Expected Due Date (EDD)

Step1: Arrange the due date from the least to the highest with their corresponding Job in that order

Job	Due Date
A	16
B	20
D	20
C	25
E	40

Step2: Put their corresponding Processing time in the middle

Job	Process Time	Due Date
A	9	16
B	7	20
D	8	20
C	5	25
E	6	40
	35	121

Step3: Cummulate the Process time & Put the value between the Process Time & due dat and tag it Flow Proces or Flow time

Job	Process Time	Flow Time	Due Date
A	9	9	16
B	7	16	20
D	8	24	20
C	5	29	25
E	6	35	40
	35	113	

Seatency

the facility

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Step 4: Determine the late Jobs by checking Due date with Flow time

Job	Process time	Flow time	Dele Date	Late Job
A	9	9	16	0
B	7	16	20	0
D	8	24	20	4
C	5	29	25	4
E	6	35	40	0
	35	113		8

Using Expected due date

1) Total completion time is 35 days
2) Average completion time = $\frac{\text{Total flow time}}{\text{No of Jobs}} = \frac{113}{5} = 22.6$ days
3) Average Job lateness = $\frac{\text{Total lateness}}{\text{No of Jobs}} = \frac{8}{5} = 1.6$ days
4) Average number of works @ the center = $\frac{\text{Total flow time}}{\text{Total Process time}} = \frac{113}{35} = 3.23$ Jobs
5) Tardy Job = 2 Jobs ⇒ Job D & Job C

Sequencing - Johnson's Algorithm

We apply Johnson's algorithm where there is a number of Processing time for sequencing 'n' Jobs for through 2 machines or work center and where the same process sequenced must be followed by all Jobs with shortest Processing time

Mathematical Condition to be Satisfied

Let there be 'n' numbers of Jobs and 2 machines or work Centers

1) Let i = 1 and P = n
2) Check the Process time for the least time
3) If the least time occurs on work centre 1, then Schedule K on
   i = i - 1
4) Otherwise schedule on P = P - 1

If all Jobs are scheduled STOP, otherwise go to 2

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MARKOV PROCESS

Some management systems changes with time & some of those system operate with some probability. The state of a process is determined by the present situation to decide what next stage to be and not based on the past

Introduction

Many management system changes regularly with time. Such a system can be analyzed in order to predict the next stage of the system by using MARKOV MODEL. The Markov model is a probabilistic device that operate based on probability. In order to fully understand our discussion on Markov process, there is a need to first discuss the terms that are used in markov process

To define the various term we shall first use some example of buyers switching behaviour in the purchase of 2 brands of toothpaste

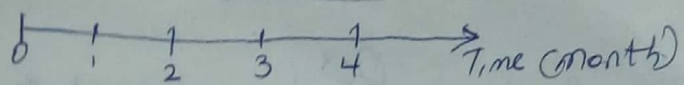
Example: A manufacturer of Hi-Glo toothpaste currently controls 45% of the market in a particular city CNB this is relating to the share of the market. Data from the previous year shows that 60% of Hi-glo customers remain loyal to Hi-Glo while the remaining 40% of Hi-glo customers switched to rival brand named Herbal toothpaste. In addition 70% of Herbal toothpaste customers remain loyal to Herbal toothpaste while the other 30% of herbal toothpaste customers switch to Hi-Glo toothpaste. Assuming that this trend continues, we can use a markov model to determine the market share of Hi-Glo toothpaste in 2 years, 3 years & over the long run

Definition of Terms in Markov Model

1) Time: This refers to a point of action. This is refers to the time the customer is purchasing either Hi-Glo toothpaste or herbal toothpaste. Suppose each customer could walk into a supermarket at the start of each month to buy either Hi-glo toothpaste or Herbal toothpaste then our time can be represented by an horizontal axis divided into month with the beginning of the axis denoted as time zero as shown below

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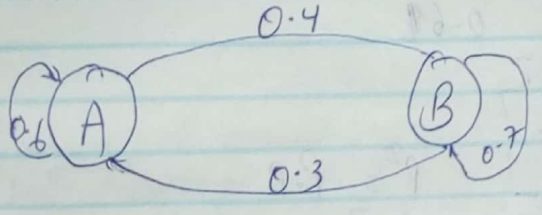
Time Axis



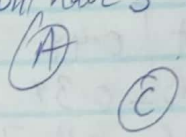
1) STATE: This simply refers to the type of toothpaste a customer is purchasing at a point in time (ie at the beginning of each point). So if the denotes the purchase of Hi-Glo toothpaste is state A, and the purchase of Herbal toothpaste in state B, then a customer is either in STATE A or STATE B in any one month. Hence A and B become the states of the system

2) TRANSITION: This is refers to the change movement of a customer from one product to another in this case from state A to B or remaining in state A itself. Thus someone can move from state A to back to state A or to state B

Transition Diagram: This is a diagram that shows the movement of a customer from one state to another.



In a situation we have 3 state we will have 3 samples like



Transition MATRIX

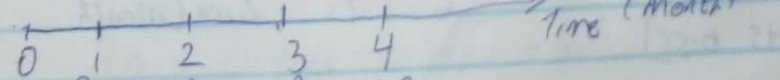
A Matrix is an array of number arrange in rows & column

A Transition Matrix is an array of Probabilities showing a switching behaviour of a customer from one state to another

$$P = \begin{array}{c} \text{THIS} \\ \text{MONTH} \end{array} \begin{array}{c} \\ A \\ B \end{array} \begin{array}{c} \text{Next month} \\ A \quad\quad B \end{array} \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

As so long as these variables remain the same, this will be what will be used to determine their future

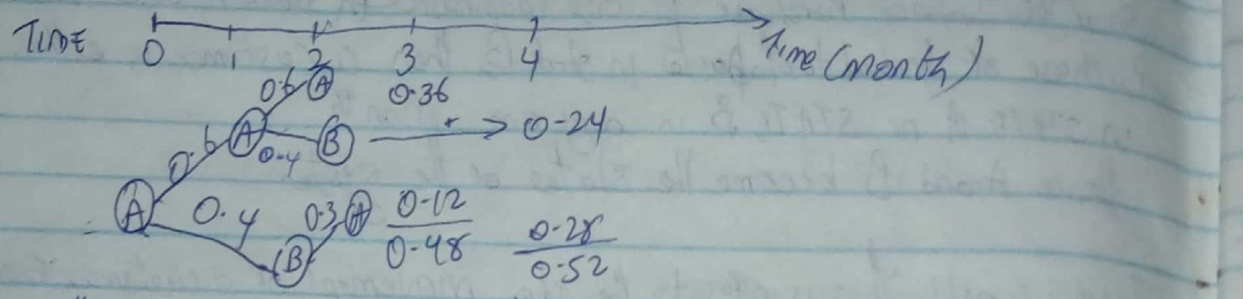
Suppose at state A (0) customer buy Purchase Hi-Glo & later switch to B. till state time 3 what it is the Probability of moving from time 3 to time 4



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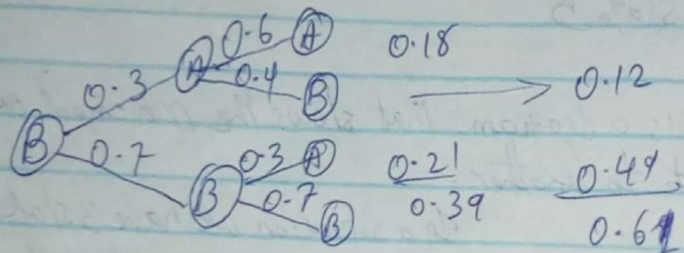
The Probability of B to A is 0.3 due to you don't consider Past event

Example if Original root start A there are many ways stated below



Thank

This method is a decision Tree Analyses



$$P^2 = \begin{matrix} & A & B \\ A & 0.48 & 0.52 \\ B & 0.39 & 0.61 \end{matrix} \quad P^2 = P \cdot P$$

$$\begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

$$= \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix}$$

P^3 in algebra is stated as $P \cdot P^2$ but in Matrix $P^2 P$ & $P P^2$ are not the same thus you can only use $P^2 \cdot P$ in Matrix

$$= \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

$$= \begin{bmatrix} 0.444 & 0.556 \\ 0.417 & 0.583 \end{bmatrix}$$

Market Share Analysis

The Market time for Hi-Gold toothPaste as start 0 is [0.45 0.55]



[0.45 0.55]

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Part

Denoting the initial market share by S_{A_0} and S_{B_0}, then

$[S_{A_0}\ S_{B_0}] = [0.45\ \ 0.55]$

Market share at Point 1 can be written as

$[S_{A_1}\ S_{B_1}] = [S_{A_0}\ S_{B_0}] \times P$ [which is the transition matrix]

i.e $[0.45\ \ 0.55]\begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$

$= [0.435\ \ 0.565]$

$[S_{A_2}\ S_{B_2}] = [S_{A_0}\ S_{B_0}]P^2$

$= [0.45\ \ 0.55]\begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix} = [0.4305\ \ 0.5695]$

$[S_{A_0}\ S_{B_0}] = [0.45\ \ 0.55]$

$[S_{A_1}\ S_{B_1}] = [0.435\ \ 0.565]$

$[S_{A_2}\ S_{B_2}] = [0.4305\ \ 0.5695]$

For $[S_{A_3}\ S_{B_3}] = [S_{A_0}\ S_{B_2}] \times P^3$

INVENTORY ANALYSIS

Suppose a supermarket in the city ~~sells~~ stop the 2 brands of Product & sells 2000 tubes of both Product every month. What number of unit of each Product should be stocked in month 1, 2 or 3

for each state in time the market share determine what will be kept

eg $[S_{A_0}\ S_{B_0}] = [0.45\ \ 0.55] = [0.45 \times 2000\ \ \ 0.55 \times 2000]$

$\approx [900\ \ 2100]$ for $[S_{A_1}\ S_{B_1}] = [0.435 \times 200\ \ 0.565 \times 2000] =$

From here we can proceed to compute the Revenue =

The Market share X Cost Price or Profit margin

STEADY STATE OF THE SYSTEM [long-run Probabilities]

At this stage they will be a fixed state of either Hi-glo or Herbal tooth Paste will be Purchase

Denote the steady study by $\Pi = [\Pi_A\ \Pi_B]$ with $\Pi_A + \Pi_B = 1$

Then, $[\Pi_A\ \Pi_B] = [\Pi_A\ \Pi_B]\begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$

$= [0.6\Pi_A + 0.3\Pi_B \quad 0.4\Pi_A + 0.7\Pi_B]$

$\Pi_A = 0.6\Pi_A + 0.3\Pi_B \quad -①$

$\Pi_B = 0.4\Pi_A + 0.7\Pi_B \quad -②$

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Pending the initial market share by S_{A_0} and S_{B_0}, then

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Suppose a supermarket in the city sells stop the 2 brands of Product & sells 2000 tubes of both Product every month. What number of unit of each Product should be stocked in month 1, 2 or 3

For each state in time the market share determine what will be kept

eg $[S_{A_0} S_{B_0}] = [0.45\ \ 0.55] = [0.45 \times 2000\ \ \ 0.55 \times 2000]$

i.e $[900\ \ 2100]$ for $[S_{A_1} S_{B_1}] = [0.435 \times 200\ \ 0.565 \times 2000] =$

From here we can proceed to compute the Revenue =

The Market share X Cost Price or Profit Margin

STEADY STATE OF THE SYSTEM [long-run Probabilities]

At this stage they will be a fixed state of either Hi-glo or Herbal tooth paste will be Purchase

Denote the steady study by $\pi = [\pi_A\ \ \pi_B]$ with $\pi_A + \pi_B = 1$

Then, $[\pi_A\ \ \pi_B] = [\pi_A\ \ \pi_B] \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$

$= [0.6\pi_A + 0.3\pi_B \quad 0.4\pi_A + 0.7\pi_B]$

$\pi_A = 0.6\pi_A + 0.3\pi_B \quad -(1)$

$\pi_B = 0.4\pi_A + 0.7\pi_B \quad -(2)$

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NB when 2 Vector are equal it is compulsory that their corresponding vector should be equal

Since $\pi_A + \pi_B = 1 \Rightarrow \pi_A = 1 - \pi_B$ ← (2)

$1 - \pi_B = 0.6[1 - \pi_B] + 0.3\pi_B$

$1 - \pi_B = 0.6 - 0.6\pi_B + 0.3\pi_B$

$1 - \pi_B = 0.6 - 0.3\pi_B$

Collect like terms

$1 - 0.6 = \pi_B - 0.3B$

$0.4 = 0.7\pi_B$

$\pi_B = \frac{4}{7} =$

$\pi_A = 1 - \frac{4}{7} = 3/7$

At the steady stage

$$\lim_{n \to \infty} P^n \quad \begin{matrix} A \\ B \end{matrix} \begin{bmatrix} 3/7 & 4/7 \\ 3/7 & 4/7 \end{bmatrix}$$

Therefore $[\pi_A \ \pi_B] = [3/7 \ \ 4/7]$

MEAN RETURN TIME

This is to say if a customer start with a state now, how long will it take the customer to return back to that state

Mean return time for state $i = \frac{1}{\pi_i}$

Therefore the mean return time for state $A = \frac{1}{\pi_A}$

" " " " " " " $B = \frac{1}{\pi_B}$

Markov Chain

This is different from markov Process. When a system changes states in district (i.e at whole number point) is a markov Chain but when a system changes states in continously in time it is called Markov Process

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...available then choose

16/4/2019

Linear Programming

In the business world, one thing is constant that is decision making. While making decision you will undergo either of the 3

1) Deasion making under Uncertainty
2) " " " risk/stochastic
3) " " " Certainty

The factors that you need to be aware of as they undermine and give information on which of the best decision to make they are are called Decision Variables. They are of 2 types

1) Controllable Decision Variable 2) Uncontrollable Decision Variable

1) Controllable decision variable are those decision variable that are within the control or purview of the decision maker classified into 2 either the decision maker can be an individual or a group. (The decision maker can be an individual or a group depending on the authority that gives the approval). Controllable D.V sometimes are called alternatives or options

2) Uncontrollable Decision Variable are Decision variable that are outside the purview of the decision maker eg the Price of a Product is not under the control of the buyer. This can be referred to as State of nature, or events.

The uncontrollable decision variable determine the outcome of the final decision. For example if you don't have enough information about the occurrence of the uncontrollable variable than you are at the uncertainty level but you can't ascribe Probability distribution but though you don't have enough information about the occurrence of the uncontrollable variable but you can ascribe Probability distribution then you are at Risk situation

Whereas if you have detailed information about the occurrence of a situation decision variable then you are at Certainty situation.

Operative research has provided model to guide in making decision to subjectivity. In certainty situation, the model used is called Linear Programming

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Linear Programming (L.P) is a mathematical method of allocating scarce & limited resources to uses by a decision maker. It employs mathematical expressions to model or Programme a Problem situation. Sometimes it is referred to as Allocation Problem.

It is a linear objective constraint optimization technique which seeks the linear achievement of a single objectives of either maximizing or minimizing unknown variable in a model. This unknown variable are usually Profit related in a maximization Problem & cost related in a minimization Problem.

However, since the maximization Profit or the minimization cost is always constraint by real world limitation of finite resources, linear Programming Provide the user of the technique an opportunity to combine the constraining limitation of the decision environment with the interaction of the unknown values that we seek to optimize.

NB Isoobjective model in linear Programming means you either maximize or minimize this you can't achieve both at once

It should be noted that L.P technique can only be applied within all linear frameworks. In linear programming the highest power of x is one (x¹) i.e the situation whereby there is only one objective & the unknown variables are the criterion function & the constraint equation are all linear under the condition of certainty

Assignment 1

Identify at least 10 areas where linear Programme can be applied in the discipline of finance with real life situation of different industry

Terminologies of linear Programming

i) Decision Variables: A set of unknown quantities represented in every model by mathematical symbol which values are to be determined in order to either maximize or minimize a given criterion function. It is denoted by x_j where x = Decision Variable, j = no of DVs column

The activities to be embark upon determine your decision variable eg the quantity to Produce in order to maximize profit & reduce cost.

NB for x_{ij} the i will be row & j is column of DVs

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b) Objective function: The objective function is denoted by capital Z and C where Z is for maximization problem & C is for minimization problem. It is a linear function expressed as the sum of the product of contribution (or cost) coefficient with their respective decision variable. It is the unknown function that we seek to either maximize or minimize - to achieve the maximization objective of the vendor each of the DVs bring an amount that amount is called Contribution. There are measured values, are Coins.

Example

A vendor who usually sells indomie of different quantities like Normal, Suberpack, Hungry man size, B 100 150 220 & 300 are at their respective quantity and they ~~contribute 100, 150~~, they contribute i.e the value it is bringing to the business

The profit maximization will be

$$Z = 100x_1 + 150x_2 + 220x_3 + 300x_4$$

c) Technological Coefficient

These are numerical value of constraint represented by a_{ij} that express the per unit usage of right hand side ~~result~~ resource

R_i represent the number of resource(s) where $i = 1 \ldots \ldots D$

b_i represent limit of availability; i.e the quantity of your resources

eg the indomie vendor Resources could be

		b_i
R_1	Spice	b_i
R_2	Egg	30
R_3	Pepper	50g
R_4	Gas	12.5kg
R_5	Groundnut oil	25 litres
R_6	Disposable Pack	100

If he want to cook N S H B b_i

where R_1 10 15 20 25 500g

LHS RHS

$10x_1 + 15x_2 + 20x_3 + 25x_4 \leq 500$

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SEQUENCING

There are some assumption (Explain & know them) → Read the assignment

1) Linearity: Means there is constant return to scale
2) additivity: This means in order to achieve your objective you have to sum up the
3) Deterministic or certain: i.e all the information about the parameters are available eg bi
4) Proportionate: Constant return to scale: i.e increase in Put by same measure as output i.e 1↑1
5) Divisibility: Fractional value is allowed: i.e Privileged a continous value
6) Finiteness: i.e all resources are exhaustable: i.e at the beginning of the decision all the resources are known & you can't add additional resources
7) Non-negativity conditions: all values of x are not allowed to be negative - that why in L.P zero is a value - you must do your analysis before you write out the activity you want to carried out

Mathematical formation of linear Programming

1) Derived or determined decision variable eg x_{ij} are to Produce Product or sell in domie Pack where x_i stands for normal
2) Derive or determine your objective function the objective here is to achieve Profit
3) Derive or write out your constraint equation. This is determine by your resources.

NB if the Problem is maximization Problem it is ≤ your equation constraints
Minimization Problem ≥ or =

If all sign are same throughout symmetrical L.P i.e
≤ ≥ or = if not same asymmetrical L.P problem
≤ ≥
≤ or ≥ =

In formulating the Problem to denote whether it is less than or more than eg if he want to make Profit he must not sell more than 25 Pack

SIGN	Interpretation	Example
≥	Greater than or equal to at least minimum no of not less than	At least 25% of total no of houses built must be on the duplex style

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>	Greater than more than	More than 50% of investment portfolio must consist of shell shares
=	Equals	Production balancing for a component what you buy & what you make or a component must equal what you sales what ~~you~~ store
≤	less than or equal to, at most, maximum no of, not more than	staff can not ~~longer~~ work more than 50 hrs per week
<	Less than	staff must work less than 35 hrs per week

NB: All these terms are important only under constraint equation
First stage: state your non-negativity Phase condition
where i.e. $X_j \leq 0$ where $J = 1, 2, 3, 4$

$$x_1, x_2, x_3, x_4 \leq 0$$

NB maximize is Concave but Minimize is Convex
for example → TRF note

A company engages in Producing ~~two~~ two Products from 2 resources which are materials & labour, each unit of Product 1 will require 1 Pound of material & 6 hrs of labour. each unit of Product 2 requires 2 Pound of material & 6 hrs of labour. The Contribution to per unit of Product 1 is ₦4 while each unit of Product 2 Contributed ₦5 to Profit. The total materials available for the Production of these Products is limited to 10 Pounds while in all 36 labour hours is available.

Further, sale Product forecast as limited the Product 1 to be 4. Formulate the appropriate linear Programming Problem model for the given Problem

Solution

Activity level — Product 1 — Product 2

NB Activity level is the quantity of the product to be produced

Decision Variable NB Let activity level be represented by R

Let X_j be the quantity of Product 1 and Product 2 to be Produced where $J = 1, 2$

be on the duplex sty

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Decision Variable

Material	Product A(x_1)	Product B(x_2)	Available Resources
Material	1	2	10
labour	6	6	36
Profit(Contribution (₦) per unit	4	5	

Objective function

Maximize $Z = 4x_1 + 5x_2$.

Constraints or Subject to:

Material: $x_1 + 2x_2 \leq 10$ --- ①

Labour: $6x_1 + 6x_2 \leq 36$ --- ②

$x_1 \leq 4$ --- ③

$x_1, x_2 \geq 0$

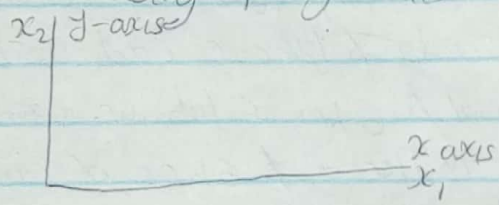
NB Non-negatively condition: It can be written as $x_j \geq 0$ where $j = 1, 2$ or $x_1, x_2 \geq 0$

Methods of Linear Method

1) Graphical method: It can only be used where you have at most two variables.

2) Simplex method is used to solve at least 2 variables

Using the grathical method



$Z = 4x_1 + 5x_2$

From equation 1

$x_1 + 2x_2 \leq 10$ ...

When/Let $x_1 = 0$

$2x_2 = 10$

$x_2 = 5$

When/let $x_2 = 0$

$x_1 = 10$

From equation 2

When $x_1 = 0$

$x_2 = 6$

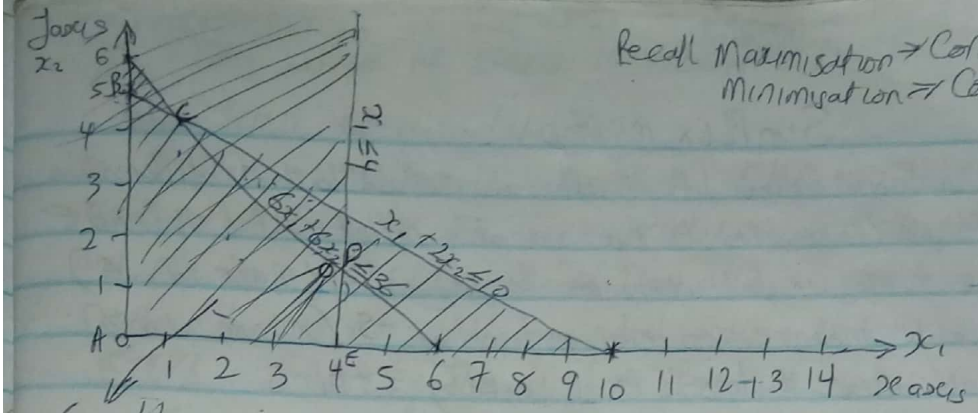
when $x_2 = 0$

$x_1 = 6$

From equation 3

$x_1 = 4$ $x_2 = 0$

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Recall Maximisation → Concave
Minimisation → Convex

Feasible region

The feasible region is the region that contains all the different shades

At Point A
$x_1 = 0, x_2 = 0$

At Point B
$x_1 = 0, x_2 = 5$

At Point C

$x_1 + 2x_2 = 10$

$6x_1 + 6x_2 = 36$ Solving Simultaneously $[-6x_1 + 6x_2 = 36]1$

$6x_1 + 12x_2 = 60$

$6x_1 + 6x_2 = 36$

$6x_2 = 24$ $\therefore x_2 = 4$

Substitute $x_2 = 4$ in equation (1) $x_1 + 2(4) = 10$

$x_1 + 8 = 10$ $\therefore x_1 = 2$

At Point D $x_1 = 4$

$6x_1 + 6x_2 = 36$

$6(4) + 6x_2 = 36 \Rightarrow 24 + 6x_2 = 36$

$6x_2 = 36 - 24 = 12 \Rightarrow x_2 = 2$

At Point E

$x_1 = 4, x_2 = 0$

Objective function Insertion

At Point A

$Z = 4(0) + 5(0) = ₦0$

At Point B

$Z = 4(0) + 5(5) = ₦25$

At Point C

$Z = 4(2) + 5(4) = ₦28$

At Point D

$Z = 4(4) + 5(2) = ₦26$

At Point E

$Z = 4(4) + 5(0) = ₦16$

Interpretation

If he wants to produce at maximum Profit he should produce 2 units of Product 1 & 4 units of Products 2 making a Profit of ₦28

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SEQUENCING

SIMPLEX METHOD

This is used in same process like graphical method but in different perspective. It uses/introduces its own variable called SLACK VARIABLE.
Slack variable means unused resources denoted by $+S_i$ (maximization $\leq$)
Surplus variable means excess resources denoted by $-S_i$ (minimization $\geq$).
Artificial variable means it doesn't exist denoted by $= +A_i - S_i$

$Z = 4x_1 + 5x_2$
Subject to $x_1 + 2x_2 + S_1 = 10$
$6x_1 + 6x_2 + S_2 = 36$
$x_1 + S_3 = 4$
$x_1, x_2, S_1, S_2, S_3 \geq 0$ (non negativity condition)

NB Simplex method use table unlike the graphical (graph)
(canonical form

Every of the variable must be present at each stage - the slack variable has the co-efficient of "1" where it is introduced and "0" in other places

$Z = 4x_1 + 5x_2 + 0S_1 + 0S_2 + 0S_3$
Subject to $x_1 + 2x_2 + S_1 + 0S_2 + 0S_3 = 10$
$6x_1 + 6x_2 + 0S_1 + S_2 + 0S_3 = 36$
$x_1 + 0x_2 + 0S_1 + 0S_2 + S_3 = 4$
$x_1, x_2, S_1, S_2, S_3 \geq 0$

C_b	C_j Basis	b_i	x_1^4	x_2^5 (PC)	S_1^0	S_2^0	S_3^0
0	S_1	10	1	2 (PE)	1	0	0
0	S_2	36	6	6	0	1	0
0	S_3	4	1	0	0	0	1
	Z_j	0	0	0	0	0	0
	$C_j - Z_j$		4	5	0	0	0

NB Pivot element or Pivot number
Pivot row
Z_j = total contribution of the current solution set (the variables in the basis)

Marginal change in the value of the objective, Z_j as a result of slack variable in the referent j-column

NB $Z_j = (\text{basis} \times b_i)$ Sum Sum of $(C_b \times b_i) = (0 \times 10) + (0 \times 36) + (0 \times 4) = 0$

* Decision Rule is found at $C_j - Z_j$
* If it is Maximization Problem all $(C_j - Z_j)$ must be either negative or zero (then you say it is optimal solution)
* If you have combination of negative & zero then it is feasible Solution
* If you have a positive value there is no solution at all - then you have to run the analysis again

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C_j - Contribution of Variables in Objective Function
C_b - Contribution of basis in objective function
b_i - limit of availability

NB: If it is a minimization problem at the $C_j - Z_j$ row, all values must be positive for an optimal solution but if you have positive and 0's it is a FEASIBLE SOLUTION and if you have negative value there is no solution and you have to run analysis again

STEP1 : Identify the highest positive value on the $C_j - Z_j$ row which is 5

Pivot column(PC): the column of the highest positive value

For Pivot row: $\frac{\text{limit of availability } (b_i)}{\text{Pivot Column } (P_c)}$

Identify minimum values ignoring negatives & Zero's

$S_1 = \frac{10}{2} = 5$ $\quad S_2 = \frac{36}{6} = 6$ $\quad S_3 = \frac{4}{0} = 0$

Pivot element: The Point/cell where Pivot row and column meet

Pivot column

C_j / C_b	Basis	b_i	x_1^4	x_2^5	S_1^0	S_2^0	S_3^0	
5	x_2	5	1/2	1	1/2	0	0	
0	S_2	6	3	0	-3	1	0	PR
0	S_3	4	1	0	0	0	1	
	Z_j	25	5/2	5	5/2	0	0	
	$C_j - Z_j$		3/2	0	-5/2	0	0	

For the new:

To get other rows use the value in your Pivot column to multiply the new row (x_2) and substract from the old value of the row you are looking for

Pivot row (PR) = $\frac{\text{limit of availability } (b_i)}{\text{Pivot Column } (P_c)}$

$S_1 = \frac{5}{1/2} = 10$ $\quad S_2 = \frac{6}{3} = 2$ $\quad S_3 = \frac{4}{1} = 4$

C_j / C_b	Basis	b_i	x_1^4	x_2^5	S_1^0	S_2^0	S_3
5	x_2	4	0	1	1	-1/6	0
4	x_1	2	1	0	-1	1/3	0
0	S_3	2	0	0	1	-1/3	1
	Z_j	28	4	5	1	1/2	0
	$C_j - Z_j$		0	0	-1	-1/2	0

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Introduction

Solution ① Solve for initial basic feasible solution

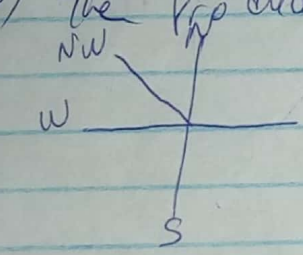
Demand / Supply	D_1	D_2	D_3	D_4	SS
S_1	a) 500 [190]	b) 200 [300]	X [500]	X [100]	700 → 200 → 0
S_2	X [700]	c) 600 [300]	d) 300 [400]	[600]	900 → 300 → 0
S_3	X [400]	X [50]	e) 400 [700]	f) 1400 [200]	1800 → 1400 → 0
DD	800 → 600	800 → 600 → 400	700 → 400	1400	3400

Using North-West Corner Method: It is suggested by George Dantzig. The steps are

1) Partition the entire cost matrix into cardinal Point
2) Locate the most North-West (maximum) Cell
3) Make an optimum allocation to that cell without violating the row & column Constraint

NB In business, it is ideal that you supply what the market demand if you supply more than demand it shows that

a) You are tying down Capital
b) The Product is not Competitive



the optimum allocation cell is the 190 cell which is D_1S_1: NW is 190

NB you should not supply more than Demand therefore you should apply 500 (DD) out of 700 (SS). For S_1D_1 because the Demand is 500

4) Evaluate the next most north west cell

Degeneracy Problem occurs in transportation model when the total allocation is not equal to row + column − 1 (one)

Degeneracy Problem Prevents from Performing 2 Critical Operation in Transportation model

1) It Prevents you from calculating the total cost of transportation
2) The solution cannot be improved upon

How to Solve Degeneracy Problem

It is resolve by introducing an artificial allocation in order to make row + Column − 1 = Total allocation

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r+c-1 = TA
3+4-1 = 6

Assume it is not equal to 6 we go to 5 that means there is degeneracy. we have to create Dummy column & add 00 like [D 00]

NB 6 represent 6 Cells i.e Artificial intelligence & the ans = 6: 6-1

Allocation	Unit Allocation	Cost Per unit	total Cost
a	500	190	95,000
b	200	300	60,000
c	600	300	189,000
d	300	400	120,000
e	400	700	280,000
f	1400	200	280,000
			1,015,000

Least Cost Method

Here, we consider the entire cost matrix & locate the most least cost cell i.e S_3D_2

	D_1	D_2	D_3	D_4	Ss
S_1	X 190	X 300	X 500	b) 400	700 a)
	d) 700			600	
S_2	200	X	d) 700	400	300
	e) 400	80	400		
S_3	300	a) 800	X	d) 200	1400 300
				700	
DD	500 200	800	700	1400	3400

A Charge occurs when 2 cells are having the same cost of transportation & even more visible.

The most least cost cell is a) S_3D_2 b) S_1D_4 c) S_3D_4 d) S_2D_3 e) S_3D_1 f) S_2D_1

⇒ To confirm if it degenerates r+c-1 ≟ 3+4-1=6 There is no degeneracy

Allocation	Unit allocation	Cost Per Unit	Total cost
a	800	80	64,000
b	700	100	79,000
c	700	200	140,000
d	700	400	280,000
e	300	400	120,000
f	200	700	140,000
			814,000

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Vogel Approximation Method

SS \ DD	D_1	D_2	D_3	D_4	SS	P_1	P_2	P_3
S_1	500	X	300	200	700	90	90	400
S_2	X	X	700	200	900	100	200	200
S_3	X	800	X	1000	1800	120	200	500
DD	500	800	700	1400	3400			
P_1	240	220	100	100				
P_2	240	-	100	100				
P_3	-	-	100	100				

Here, once you know the implication/penalty of taking wrong decision, you will be guided to make the right choice.

There are 5 Procedures

1) Identify the 2 least cost along the row & column. Then find the different which is the penalty
2) Identify the highest penalty row or column
3) Identify the least cost cell along the highest penalty & make optimum allocation to the cell
4) Repeat step 1 to 3
5) Follow the least cost Procedure on a 2 by 2 matrix

$r+c-1 = TA \Rightarrow 3+4-1=6$

Allocation	Unit Allocation	Cost Per Unit	Total cost
a	800	80	64,000
b	500	190	95,000
c	1000	200	200,000
d	200	100	20,000
e	700	400	280,000
f	200	600	120,000
			719,000

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